Sensors and meteorology
Project report
DIY ultrasonic sensor

Submitted by	ID	Tutorial Group	Team number	Supervised By
Aliaa Tarek	17006290	T-16	63	Dr. Lobna Tarek
Zeinab mahmoud	16009962	T-11		

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Project Overview

This project involves designing and implementing a DIY ultrasonic distance sensor module from scratch. The system uses a transmitter and receiver, with signal amplification and a microcontroller interface.

Arduino Nano (ATmega328P): Handles pulse generation, echo detection, and time-of-flight measurement.
Arduino Uno: Receives the measured distance from the Nano (via Serial or other communication) and displays it on the Serial Monitor or an LCD.

The system generates 40 kHz ultrasonic pulses, which are transmitted through air, reflected off objects, and detected by the receiving transducer. The received signals are amplified using LM358 operational amplifiers, and a MAX232 IC is used to boost the transmitter voltage (~30V peak-to-peak) to increase the sensor range.

Project Objectives

1. Build a fully DIY ultrasonic sensor module.
2. Generate 40 kHz ultrasonic pulses.
3. Amplify the received echo signal for accurate detection.
4. Measure time-of-flight using Arduino Nano to calculate distance.
5. Send distance data to Arduino Uno for display.
6. Display measured distance in real time on the Serial Monitor .
7. Prototype on a breadboard .

System Functionality

1. Triggering**:** Arduino Nano detects a trigger input and generates eight cycles of 40 kHz ultrasonic pulses.

2. Transmission**:** MAX232 boosts the signal, and the transmitter speaker emits ultrasonic waves.

3. Reflection**:** Waves reflect off objects.

4. Reception**:** Receiving transducer converts echoes into weak electrical signals.

5. Amplification**:** LM358 amplifier stages filter and amplify the signal to produce a clean digital pulse.

6. Time-of-Flight Measurement: Arduino Nano measures the time delay between transmission and reception.

7. Distance Calculation: Using the formula:

   Distance_cm = Time_us × 0.01715

8. Data Transmission: Arduino Nano sends the distance data to Arduino Uno.

9. Display: Arduino Uno prints the distance to the Serial Monitor or displays it on an LCD.

Implementation Plan

1. Component Selection: 40 kHz ultrasonic transmitter and receiver, LM358 op-amps, MAX232, Arduino Nano and Uno, resistors, capacitors, potentiometers, and breadboard.

2. Transmitter Assembly: transistor-driven 40 kHz pulse circuit will be built, boosted by MAX232 IC.

3. Receiver Assembly: Connect receiving transducer to LM358 amplifier stages; adjust thresholds via potentiometers.

4. Arduino Nano Integration: Connect trigger and echo pins; program interrupts to generate bursts and measure echo pulse width.

5. Distance Calculation: Arduino Nano computes the distance and sends it to Arduino Uno.

6. Arduino Uno Integration: Receive distance data via Serial and display on Serial Monitor or LCD.

7. Breadboard Testing: Verify signal generation, echo detection, amplifier function, and distance display.

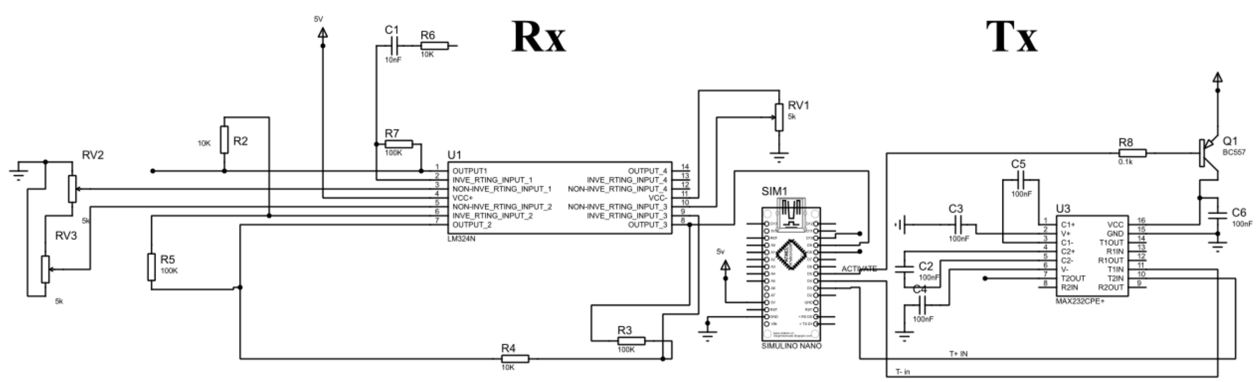
8. Calibration and Validation: Adjust amplifier and MAX232 voltage to optimize detection and validate measurements against known distances.

3. Electrical Circuit Diagram

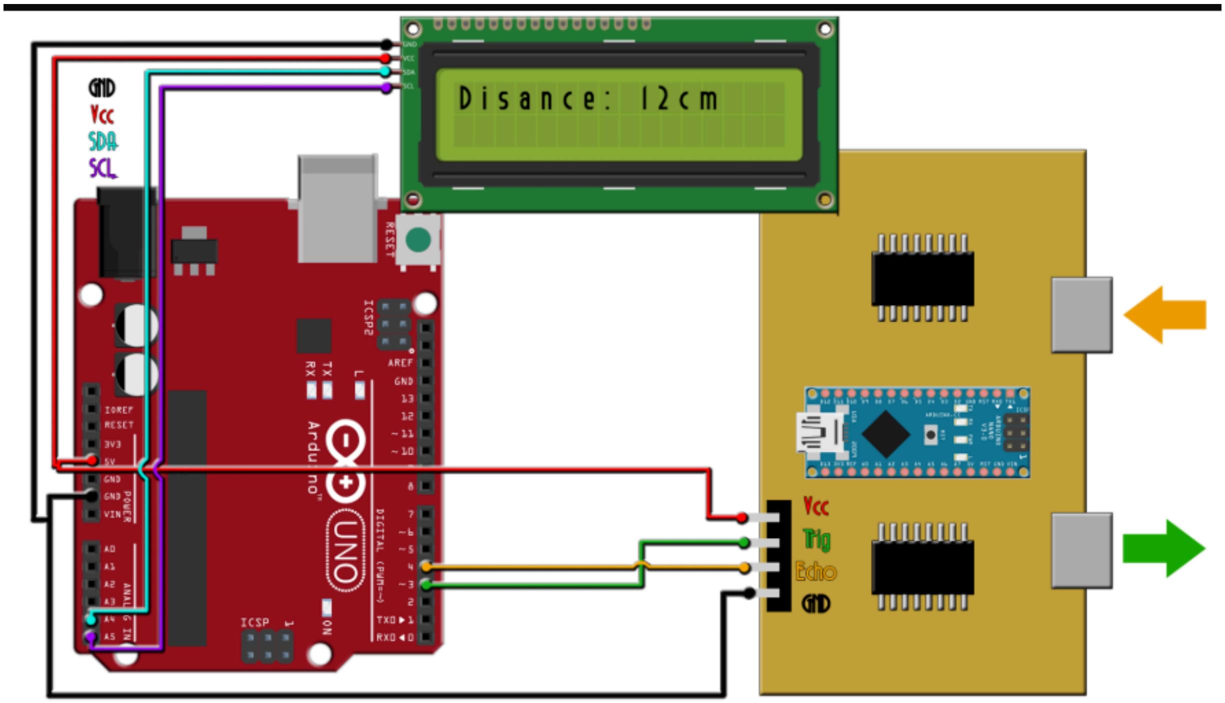
Key Points:

- Trigger and echo pins on Arduino Nano connected to amplifier and transmitter circuits.
- MAX232 IC and PNP transistor for transmitter voltage boost.
- LM358 amplifiers for receiver signal conditioning.
- Arduino Nano sends distance via Serial to Arduino Uno.
- Arduino Uno prints distance on Serial Monitor or optional LCD.
- All power and ground lines are properly connected.

Schematic made with proteus



Original schematic showing the final connection of the sensor with the arduino uno



4. Bill of materials (BOM)

Category	Reference	Quantity	Component	Value	Estimated Price (EGP)	Availability
Capacitors	C1	1	Ceramic Capacitor	10 nF	0.25	RAM Electronics
Capacitors	C2, C3, C4, C5, C6	5	Ceramic Capacitor	100 nF	1.25 (0.25 × 5)	RAM Electronics
Resistors	R2, R4, R6	3	Fixed Resistor	10kΩ	1	RAM Electronics
Resistors	R3, R5, R7	3	Fixed Resistor	100kΩ	1	RAM Electronics
Resistors	R8	1	Fixed Resistor	0.1kΩ (100Ω)	1	RAM Electronics
Integrated Circuits	U1	1	LM324N Quad Op-Amp	—	7	RAM
Integrated Circuits	U3	1	MAX232CPE +	—	28	UGE
Transistors	Q1	1	BC557 (PNP Transistor)	—	1	Future Electronics
Miscellaneous	RV1, RV2, RV3	3	Potentiometer	5kΩ	18 (6 × 3)	RAM Electronics

Arduino Nano	SIM1	1	Board (ATmega328P)	—	250	RAM Electronics
—	—	1	Arduino Uno (optional)	—	300	RAM Electronics
—	—	1	Ultrasonic TX Transducer	40 kHz	35	RAM Electronics
—	—	1	Ultrasonic RX Transducer	40 kHz	35	RAM Electronics
—	—	1	LM358 Operational Amplifier	—	10	RAM Electronics
—	—	2	Jumper Wires Set	—	70 (35 × 2)	RAM
—	—	1	Breadboard	—	35	RAM Electronics
Total Estimated Cost				~850 EGP		